TABLE OF CONTENTS

SL no	Content	Page no
1	**INTRODUCTION**	1-7
1.1	Brief introduction about the project	1
1.2	Project Background	2
1.3	Existing System	2
1.4	Proposed System	3
1.5	Basic Design principle of the project	4
1.6	Basic Architecture of the project	6
2	**SYSTEM STUDY AND ANALYSIS**	9 -19
2.1	Drawbacks of the Existing System	9
2.2	Problem Statement	10
2.3	Feasibility Study	11
2.4	Project Planning	13
2.5	Requirement Specification	15
3	**SYSTEM WORK FLOW**	16
3.1	Design Methodology	16
3.2	UML diagrams	18
3.3	Use case design	19
4	**AGILE TECHNOLOGY OVERVIEW**	20-26

SL no	Content	Page no
4.1	Introduction to Scrum	20
4.2	Principles or methodology used	21
4.3	Product Backlog	22
4.4	Sprint Planner	23
4.5	Ideal Burn down chart	23
5	**SYSTEM DESIGN**	27-30
5.1	Technology / framework [MVC]	27
5.2	Module Description	29
6	**CODING**	31-38
6.1	Coding standards used in the project	31
6.2	Project coding	34
7	**TESTING**	39-40
7.1	Unit Testing	39
8	**IMPLEMENTATION**	41-43
8.1	Implementation method used in Project	41
9	**BUSINESS PERSPECTIVE**	44-46
10	**FUTURE SCOPE OR ENHANCEMENTS**	47-49
11	**APPENDIX**	50-54
11.1	Git History [sprint wise]	50

SL no	Content	Page no
11.2	Screen Shots of Forms, Reports, Diagrams	52
11.3	Burn down charts	54
12	**REFERENCES / BIBLIOGRAPGHY**	55-56

1. INTRODUCTION

1.1 Brief introduction about the project

Road2Career is an AI-powered personalized career mentoring and roadmap generation platform developed to help students choose a suitable career and prepare for their future professional goals. Nowadays, many career opportunities are available in different fields, such as software development, data science, cybersecurity, and cloud computing. However, students often find it difficult to identify the right career path because they may not have sufficient knowledge about the skills and qualifications required for different professions. Road2Career aims to address this problem by providing students with proper career guidance and a clear direction for their professional development.

Traditional career guidance platforms often provide general career suggestions, which may not fully meet the individual needs of students. Students may also find it difficult to understand their existing skills, identify missing skills, and decide what they should learn next. Road2Career provides a centralized platform that combines career selection, skill-gap analysis, personalized roadmap generation, learning resources, and progress tracking. This helps students understand their current abilities and plan their learning activities according to their career goals.

The system allows students to register, log in, manage their profiles, and add their existing technical skills. Based on the student's selected career goal and available skill information, the system analyzes the skills required for the chosen career and compares them with the student's current skills. This process identifies skill gaps and highlights the areas that need improvement. By understanding these gaps, students can focus on learning the most relevant skills instead of spending time on unrelated topics.

One of the main features of Road2Career is its personalized career roadmap generator. It provides students with a structured learning path based on their selected career goals and current skill levels. The system also includes a weekly learning planner that organizes learning activities into manageable tasks. In addition, it provides project suggestions and learning resources to help students improve their technical knowledge and gain practical experience. These features encourage students to follow a systematic learning process and develop the skills required for their desired careers.

Road2Career also includes progress tracking and career-readiness features that help students monitor their learning journey. Students can review their completed activities, track their progress, and identify areas that require further attention. The system uses React for the frontend, Node.js and Express.js for backend development, and PostgreSQL for database management. Its recommendation service uses rule-based and heuristic logic to analyze skills and generate career guidance. These technologies support an interactive user interface, organized data management, and the delivery of personalized career-planning features.

Overall, Road2Career aims to make career planning easier, more organized, and more accessible for students. By integrating career guidance, skill-gap analysis, personalized roadmaps, weekly learning plans, project recommendations, and progress tracking into a single platform, it helps students make better-informed decisions about their professional future. The system encourages students to identify their strengths, improve their weaknesses, and work consistently toward their career goals. Therefore, Road2Career serves as a supportive tool for students who want to develop relevant skills, improve their career readiness, and prepare themselves for future employment opportunities.

1.1 Project Background

In today's rapidly evolving digital world, choosing a suitable career has become an important challenge for students. Many career opportunities are available in fields such as software development, data science, cybersecurity, cloud computing, and other technology-related areas. However, students often find it difficult to identify the right career path based on their interests, existing skills, and abilities. They may also lack proper guidance about the skills they need to develop, the projects they should complete, and the learning resources they should follow to achieve their career goals. This creates a gap between students' current abilities and the requirements of their desired careers.

Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator aims to address these challenges by providing a personalized career guidance platform for students. The system helps students select their career goals, manage their profiles, and record their existing skills. Based on the selected career path and the student's current skills, the platform performs skill-gap analysis to identify missing skills and areas that require improvement. This enables students to understand their current position and make informed decisions about their future career development.

The platform's main feature is its personalized career roadmap generator, which provides a structured learning path according to the student's selected career goal and skill level. It also includes a weekly learning planner that organizes learning activities into manageable tasks. In addition, the system provides project suggestions and learning resources to support practical learning and improve technical knowledge. These features help students follow a clear learning path instead of depending on random tutorials or unplanned learning activities.

Road2Career also includes progress tracking and career-readiness features that allow students to monitor their learning activities and evaluate their preparation for their desired career. By combining career guidance, skill-gap analysis, roadmap generation, weekly planning, project recommendations, and progress monitoring in a single platform, Road2Career aims to make career planning more systematic, accessible, and effective. Overall, the project supports students in identifying their strengths, improving their weaknesses, and preparing for future employment opportunities.

1.3 Existing System:

In the current digital learning environment, several online platforms provide career guidance, skill development courses, and learning resources. Platforms such as Coursera, Udemy, and other online learning websites offer educational content in various technical and professional fields. Career guidance websites and job platforms also provide information about career opportunities and required qualifications. However, students may need to use multiple platforms to identify a suitable career, learn the required skills, find practical projects, and monitor their overall preparation.

Existing career guidance systems often provide general career suggestions based on a user's interests, educational background, or responses to assessments. Although these recommendations can be helpful, they may not provide a complete, personalized learning path that considers the student's current skill level and identifies the specific skills that need improvement. As a result, students may find it difficult to determine which skills they should learn first and how to organize their learning activities effectively.

Another limitation of existing systems is that career selection, skill-gap analysis, learning

resources, and progress tracking are often handled separately. Students may need to switch between different applications to access courses, practice technical skills, work on projects, and evaluate their progress. This can make career preparation time-consuming and difficult to manage. Furthermore, without a structured weekly learning plan, students may struggle to maintain consistency and complete the activities required to achieve their career goals.

Existing platforms may also provide limited guidance on how students can move from their current skill level to the requirements of a specific career. Students need more than a list of recommended courses; they require a clear roadmap, suitable project suggestions, and a way to monitor their progress. Without these integrated features, students may face difficulties in identifying their weaknesses, prioritizing learning tasks, and understanding their career readiness.

Overall, existing systems provide useful educational resources and career information, but students may still lack a unified platform that connects career selection with personalized skill analysis, structured roadmaps, weekly learning plans, project recommendations, and progress monitoring. Road2Career aims to address these limitations by integrating these features into a single platform, helping students plan their learning journey, develop relevant skills, and work systematically toward their desired careers.

1.4 Proposed System:

Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator is designed to overcome the limitations of existing career guidance systems by providing personalized career recommendations, skill-gap analysis, and structured learning plans. The system helps students identify suitable career paths, understand the skills required for their desired careers, and follow a clear roadmap to achieve their professional goals. The key features and functionalities of the proposed system are as follows:

1. **Career Goal Selection:**
    - Allows users to exchange skills with one another instead of paying for courses.
    - Allows students to select their desired career path based on their interests and professional goals.
    - Provides relevant career guidance to help students understand the requirements of their chosen career.
    - Helps students plan their learning activities according to their career objectives.
2. **User Registration and Profile Management:**
    - Allows students to register and log in securely to the platform.
    - Enables students to create and update their profiles with relevant personal and educational information.
    - Allows students to add, edit, and delete their existing skills to maintain an updated skill profile..
3. **Skill-Gap Analysis:**
    - Analyzes the student's existing skills based on the selected career goal.
    - Compares the student's current skills with the skills required for the desired career.
    - Identifies missing skills and areas that require improvement.

- Helps students prioritize the skills they need to develop.

4. **Personalized Career Roadmap Generation:**
   - Generates a structured career roadmap based on the student's selected career path and current skill level.
   - Organizes the required skills and learning activities into a step-by-step learning path.
   - Provides guidance on which skills to learn first and which skills need further practice.
   - Allows students to follow a clear direction toward their career goals
5. **Weekly Learning Planner**
   - Generates a weekly learning plan to organize learning activities and tasks.
   - Helps students divide their learning goals into manageable activities.
   - Encourages consistent learning and effective time management.
   - Supports students in following their personalized career roadmap.
6. **Project and Learning Resource Recommendations:**
   - - Provides project suggestions related to the student's selected career path.
   - Helps students gain practical experience by applying their technical knowledge.
   - Provides relevant learning resources to support skill development.
   - Encourages students to improve their problem-solving abilities through practical activities.
7. **Progress Tracking and Career Readiness**
   - Allows students to monitor their learning activities and completed tasks.
   - Provides progress information to help students understand their learning development.
   - Calculates a career-readiness score based on the criteria implemented in the system.
   - Helps students identify areas that require further improvement and track their preparation for the desired career.
8. **Roadmap History Management**
   - Maintains records of previously generated career roadmaps.
   - Allows students to review their earlier roadmaps and refer to their learning plans.
   - Helps students keep track of their career-planning activities over time.
9. **Admin and Management Tools**
   - Provides administrative functionalities to manage relevant user and platform information.
   - Supports the management of career-related data and other system information.
   - Helps maintain the organized operation of the platform.
10. **Personalized and Accessible Career Guidance**
    - Combines career selection, skill-gap analysis, roadmap generation, learning resources, and progress tracking in a single platform.
    - Reduces the need for students to depend on multiple applications for career planning.

- Provides structured guidance based on the student's current skills and selected career goals.
- Encourages students to improve their skills systematically and prepare for future employment opportunities.

1.4 Basic Design principle of the project

Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator is designed based on fundamental software engineering principles to ensure usability, security, reliability, and maintainability. The key design principles followed in this project are:

1. **Modularity**
   - The system is divided into independent modules such as user management, skill-gap analysis, career roadmap generation, weekly learning planner, and progress tracking.
   - Each module can be developed, tested, and maintained separately, making the system easier to manage and update.
2. **User-Cantered Design**
   - The interface is designed to be simple, user-friendly, and easy for students to navigate.
   - Clear navigation and organized information help students access career guidance and learning activities efficiently.
3. **Reusability**
   - Common functions and interface components are reused across different modules to reduce code duplication.
   - Reusable components simplify development and make future modifications easier.
4. **Scalability**
   - The system is designed to support future additions of career paths, skills, learning resources, and user functionalities.
   - React, Node.js, Express.js, and PostgreSQL provide a structured foundation for future system expansion
5. **Security and Privacy**
   - JWT authentication is used to verify users and protect access to authorized resources.
   - User information is managed through backend validation and controlled database operations.operations.
6. **Reliability**
   - Input validation and error-handling mechanisms help manage invalid requests and unexpected errors.
   - Structured database management supports consistent storage and retrieval of student and career-related information.
7. **Maintainability**
   - The code is organized into separate frontend components, backend routes, controllers, and service modules.

- This modular structure makes debugging, testing, and updating the application easier.

8. **Personalization**
   - Career guidance is generated based on the student's selected career goal and recorded skills.
   - Skill-gap analysis and personalized roadmaps help students focus on the learning activities relevant to their goals.
9. **Progress Monitoring**
   - The system allows students to track their learning activities and review their progress toward career goals.
   - Career-readiness information helps students identify areas that require further improvement.
10. **Platform Independence**
    - The web-based application can be accessed through supported web browsers on different devices.
    - The responsive interface helps students use career-planning features across different screen sizes.

1.5 Basic Architecture of the project

Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator follows a three-tier architecture, which ensures organized development, efficient data management, and easy maintenance. The architecture consists of three main layers: the Presentation Layer, Application Layer, and Database Layer. Each layer performs specific functions and communicates with the other layers to provide personalized career guidance to students.

1. Presentation Layer (Frontend)

- TDeveloped using React.js, Vite, and Tailwind CSS.
- Provides a user-friendly interface for students and administrators.
- Supports registration, login, profile management, and skill management.
- Displays career roadmaps, skill-gap analysis, and weekly learning plans.
- Provides progress tracking and career-readiness information.

Key Components:

- Student Dashboard
- Profile and Skill Management
- Career Goal Selection
- Career Roadmap
- Weekly Learning Planner
- Progress Dashboard
- Admin Dashboard

2. Application Layer (Backend)

- Developed using Node.js and Express.js.
- Processes user requests through REST APIs.
- Handles authentication using JWT.
- Performs skill-gap analysis based on career requirements.
- Generates personalized career roadmaps and weekly learning plans.
- Manages projects, learning resources, and progress tracking.

Key Functions:

- User Authentication
- Skill-Gap Analysis
- Career Roadmap Generation
- Weekly Plan Generation
- Project and Resource Management
- Progress Tracking
- Roadmap History Management

3. Database Layer (Backend Storage)

- Uses PostgreSQL for structured data storage.
- Stores student information and profile details.
- Maintains skills and career requirements.
- Stores career roadmaps and weekly learning plans.
- Maintains project resources and progress records.
- Supports efficient data retrieval and updates.

Stored Data Includes:

- User Information
- Skills and Career Goals
- Career Roadmaps
- Weekly Learning Plans
- Projects and Learning Resources
- Progress Records
- Roadmap History

4. Supporting Services

- JWT: Provides authentication and access control.
- Axios: Connects the frontend with backend APIs.

- AI Recommendation Service: Uses rule-based and heuristic logic for
- skill analysis and roadmap generation.
- Vite: Supports frontend development and API proxy configuration.
- Winston: Supports application logging.
- Multer: Handles file uploads where required.

Overall Working Flow

1. The student interacts with the React frontend.
2. The frontend sends requests to the Express.js backend through REST APIs.
3. The backend validates requests and processes application logic.
4. The backend communicates with PostgreSQL to retrieve or store data.
5. The recommendation service generates career guidance based on available skill and career information.
6. The backend returns the results to the frontend for display.

2. SYSTEM STUDY AND ANALYSIS

2.1 Drawbacks of the Existing System:

The existing career guidance and online learning platforms provide career information, courses, and educational resources. However, students may face difficulties in selecting suitable careers, identifying skill gaps, and following a structured learning path. The main drawbacks are as follows:

1. **Generic Career Recommendations**
    - Many existing systems provide general career suggestions based on interests or educational backgrounds.
    - Students may not receive sufficiently personalized guidance based on their current skills and career goal
2. **Lack of Effective Skill-Gap Analysis**
    - Existing platforms may not clearly compare students' current skills with the skills required for their desired careers.
    - Students may struggle to identify their weaknesses and determine which skills they need to improve.
3. **Absence of Structured Career Roadmaps**
    - Many platforms provide courses and career information without a complete, step-by-step learning roadmap.
    - Students may find it difficult to decide what to learn first and how to progress toward their career goals
4. **Lack of Personalized Learning Plans**
    - Learning activities are often followed without a clear plan based on individual skill levels.
    - Students may struggle to organize their learning activities and maintain consistency.
5. **Scattered Learning Resources**
    - Career guidance, learning materials, and project suggestions may be available across different platforms.
    - Students need to switch between multiple websites or applications to access the resources required for career preparation..
6. **Limited Practical Project Guidance**
    - Some platforms focus mainly on theoretical learning and course completion.
    - Students may receive limited guidance on selecting projects that help develop skills relevant to their chosen careers.
7. **Inefficient Progress Monitoring**
    - Students may not have a centralized system to track their learning activities and completed tasks.

- It becomes difficult to identify areas that require improvement and measure progress toward career goals.

8. **Lack of Career-Readiness Assessment**
    - Students may find it difficult to understand how well they are prepared for their desired careers.
    - Without a structured readiness assessment, they may not know which skills require further development
9. **Difficulty in Managing Career Plans**
    - Students may find it challenging to maintain and review their career plans over time.
    - The absence of easily accessible roadmap history can make it difficult to refer to previously generated learning paths.
10. **Lack of an Integrated Career Guidance Platform**
    - Existing systems may provide career guidance, courses, skill development, and progress tracking as separate services.
    - Students need a unified platform that combines these functions to plan and manage their career development effectively.

2.2 Problem Statement

In today's rapidly changing technological world, students face several challenges in choosing a suitable career and preparing for future employment opportunities. Many career options are available in fields such as software development, data science, cybersecurity, and cloud computing. However, students often lack proper guidance to identify the career path that matches their interests, existing skills, and abilities. They may also find it difficult to understand the technical skills, practical knowledge, and learning activities required to achieve their desired career goals.

Another major problem with existing career guidance systems is the lack of personalized skill-gap analysis and structured learning plans. Many platforms provide general career recommendations or educational resources without clearly identifying the difference between a student's current skills and the requirements of a selected career. As a result, students may struggle to decide which skills to learn first, which projects to complete, and how to organize their learning activities effectively.

Furthermore, career planning, learning resources, project suggestions, and progress tracking are often available through separate platforms. This makes it difficult for students to manage their learning journey in one place. Students may also lack a clear method to monitor their learning progress and understand their level of preparation for their desired careers.

Therefore, there is a need for an integrated career guidance platform that provides personalized career recommendations, skill-gap analysis, career roadmap generation, weekly learning plans, project suggestions, and progress monitoring. Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator aims to address these challenges by providing a centralized platform that helps students identify their skill gaps, follow structured learning paths, improve their technical abilities, and work systematically toward their career goals.

2.3 Feasibility Study:

A feasibility study is conducted to determine whether the proposed Road2Career system is practical, achievable, and beneficial in terms of technology, cost, operation, development time, and user requirements. The following types of feasibility have been analyzed.

1. Technical Feasibility

- The project is technically feasible because it uses established web development technologies.
- Frontend: React.js, Vite, and Tailwind CSS are used to develop an interactive and responsive user interface.
- Backend: Node.js and Express.js handle application logic, authentication, and API requests.
- Database: PostgreSQL stores structured information about users, skills, career goals, roadmaps, and learning progress.
- Authentication: JWT is used for user authentication and access control.
- Recommendation Service: Rule-based and heuristic logic is used to analyze skill gaps and generate personalized career roadmaps and weekly learning plans.
- These technologies provide a suitable foundation for developing and maintaining the system.

2. Economic Feasibility

- The project uses open-source technologies such as React.js, Node.js, Express.js, and PostgreSQL, reducing software licensing costs.
- The system can be developed and tested using commonly available development tools.
- The use of existing libraries and frameworks reduces development effort and cost.
- Hosting and deployment expenses depend on the selected hosting providers and usage requirements.
- Overall, the project is economically feasible for an academic project because its core development technologies are available without mandatory licensing fees.

3. Operational Feasibility

- The system is designed to provide a simple and user-friendly interface for students and administrators.
- Students can manage their profiles, record their skills, and select their career goals.
- The platform provides skill-gap analysis, personalized roadmaps, weekly learning plans, and learning resources.
- Progress tracking helps students review completed activities and identify areas for improvement.
- The integrated features make career planning and learning management easier to organize in a single platform..

4. Time Feasibility

- The project can be developed incrementally by dividing it into separate modules.
- Initial development can focus on user registration, login, profile management, and skill management.
- Subsequent stages can include career goal selection, skill-gap analysis, and personalized roadmap generation.
- Additional modules can cover weekly learning plans, project recommendations, learning resources, and progress tracking.
- A modular development approach supports easier testing, debugging, and integration.

5. User Feasibility

- The platform is primarily designed for students who need guidance in selecting and preparing for a career.
- Its organized interface helps users access career-planning features without navigating multiple separate platforms.
- Personalized roadmaps help students understand the learning steps required for their selected career goals.
- Skill-gap analysis helps students identify the skills they need to improve.
- Progress monitoring encourages students to follow their learning plans and evaluate their development over time..

2.4 Project Planning:

1. Overview

Project planning for Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator defines the development phases, timeline, and resource allocation required to build the system. It ensures systematic development from requirement analysis to deployment while maintaining quality and efficiency. The project is organized into six sprints covering student management, skill-gap analysis, career roadmap generation, learning resources, progress tracking, and administration.

2. Project Phases

Phase	**Description**	**Deliverables**
1. Requirement Analysis	Identify user requirements and define the project scope, objectives, and main features.	Software Requirement Specification (SRS)
2. System Design	Design the system architecture, database schema, user interface, and API structure.	System Design Document, ER Diagram
3. Development	Develop frontend and backend modules, including authentication, skill analysis, roadmaps, and progress tracking.	Functional Prototype / MVP
4. Integration & Testing	Integrate modules and conduct functional, integration, and system testing.	Test Cases & Test Report
5. Deployment	Configure the application, backend services, and database for deployment.	Deployed Application
6. Maintenance & Evaluation	Identify and fix errors, evaluate system performance, and improve features when required.	Bug Reports and Maintenance Records

3. Resource Planning

- Human Resources:
  - Frontend Developer — React.js,HTML,CSS,JavaScript.
  - Backend Developer — Node.js and Express.js.
  - Database Developer — PostgreSQL database design and management
  - UI/UX Designer — user interface design and usability.
  - Tester — testing, debugging, and validation.
  - Project Coordinator — planning and monitoring project progress.

- Technical Resources:
  - Development tools: Visual Studio Code,Web browser.
  - Frontend technologies: React.js,HTML,CSS,JavaScript.
  - Backend technologies: Node.js and Express.js.
  - Database: PostgreSQL.
  - Authentication: JSON Web Token (JWT).
  - Supporting libraries: Axios, Winston, and Multer.
- Financial Resources:
  - Open-source technologies help reduce software licensing costs.
  - Development can be performed using existing computer systems and free development tools.
  - Additional expenses may include hosting, domain registration, and deployment services, depending on the selected providers.

4. Time Schedule (Tentative)

Activity	**Duration**
Requirement Analysis	1 week
System Design and Database Design	2 weeks
Sprint 1 — Project Setup and Student Module	8 hours
Sprint 2 — Career Goal and Skill-Gap Analysis	8 hours
Sprint 3 — Career Roadmap Generation	8 hours
Sprint 4 — Projects and Learning Resources	8 hours
Sprint 5 — Progress Tracking	8 hours
Sprint 6 — Admin and Final Integration	8 hours
Testing, Documentation and Deployment	Ongoing

5. Risk Management

Risk	Impact	Mitigation
Database Connection Failure	High	Check database configuration and maintain backups.
Authentication Vulnerabilities	High	Validate JWTs, protect passwords, and enforce access control.
Incorrect Skill-Gap Analysis	Medium	Test career requirements and skill comparison logic.
Roadmap Generation Errors	Medium	Validate roadmap output and handle missing or invalid data.
Development Delays	Medium	Follow sprint planning and prioritize essential features.
Data Loss	High	Perform regular database backups.
API Communication Errors	Medium	Validate requests and implement proper error handling.

6. Expected Outcomes

- A functional web application for personalized career guidance.
- Secure student registration, login, and profile management.
- Career goal selection and skill-gap analysis.
- Personalized career roadmaps and weekly learning plans.
- Project suggestions and learning resource recommendations.
- Progress tracking and career-readiness information.
- An administrative module for managing career-related data.
- A structured and maintainable application that can be extended with additional features in the future..

2.5 Requirement Specification

Category	Specification
Processor	12th Gen Intel(R) Core(TM) i5-1240P (1.70 GHz)

Category	**Specification**
RAM	8 GB or higher
Storage	500 GB SSD or higher
Operating System	Windows 10/11, Ubuntu, or macOS
Network Requirement	Stable Internet Connection for API and Live Session Connectivity
Programming Language	JavaScript (React Frontend, Node.js Backend), SQL
Frameworks	React.js (Frontend), Express.js (Backend)
Database	PostgreSQL (Local Deployment)
Development Tools / IDE	Visual Studio Code
Browser	Latest versions of Chrome, Edge, or Firefox
Version Control	Git & GitHub for collaborative version tracking

3. SYSTEM WORK FLOW

3.1 Design Methodology

1. Overview:

The Design Methodology defines the structured approach adopted to develop Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator. This methodology ensures that the system is modular, user-friendly, reliable, and maintainable. It focuses on transforming student requirements into a well-defined system architecture that supports personalized career guidance and continuous learning.

The system is designed to help students select suitable career goals, identify skill gaps, generate personalized career roadmaps, discover learning resources, and monitor their learning progress through a centralized web application.

2. Design Approach:

The project follows a Modular and Layered Architecture, allowing each module to function independently while remaining interconnected. This approach improves flexibility, simplifies debugging, and supports future enhancements.

The design is divided into the following layers:

a) Presentation Layer (Frontend)

- Developed using React.js and Vite.
- Responsible for user interaction and visual presentation.
- Includes interfaces for student registration, login, profile management, career goal selection, skill assessment, roadmap generation, and progress tracking.
- Uses Tailwind CSS to create responsive and user-friendly interfaces.
- Uses React Router DOM for navigation between pages.

b) Application Layer (Backend)

- Developed using Node.js and Express.js.
- Handles business logic, API requests, authentication, and data processing.
- Manages career goal selection, skill-gap analysis, personalized roadmap generation, weekly learning plans, and progress tracking.
- Uses JWT for authentication and authorization.
- Uses a rule-based and heuristic recommendation service to analyze skills and generate career roadmaps.

c) Database Layer

- Implemented using PostgreSQL.
- Stores student details, profiles, skills, career goals, roadmaps, learning plans, projects, resources, and progress information.
- Supports data retrieval, updating, and storage.
- Maintains relationships between different entities to ensure consistent data management.

d) Integration Layer

- Connects the frontend, backend, database, and supporting services.
- Uses Axios for communication between the frontend and backend APIs.
- Uses JWT to support secure access to protected resources.
- Integrates the recommendation service for skill-gap analysis and roadmap generation.
- Uses supporting libraries for logging, security, notifications, and interface visualization where implemented.

3. Design Process:

The design process follows a top-down approach, beginning with the overall system architecture and progressing to detailed module design.

Phase	**Description**
Requirement Analysis	Identifying student needs, system features, and project requirements.
System Design	Preparing the system architecture, data flow, and database schema.
Module Design	Defining modules such as authentication, skill-gap analysis,roadmap generation, and progress tracking.
Interface Design	Designing user-friendly interfaces for students and administrators.
Database Design	Structuring PostgreSQL tables and relationships for efficient data storage and retrieval.
Integration and Testing	Connecting frontend, backend, and database components and checking their functionality.

4. Design Models Used:

1. **Use Case Diagram:** – Illustrates interactions between students, administrators, and the system.
2. **Data Flow Diagram (DFD):** – Represents how data moves through registration, skill assessment, career analysis, roadmap generation, and progress tracking.
3. **Entity-Relationship (ER) Diagram:** – Defines relationships between database entities such as users, skills, career goals, roadmaps, learning resources, and progress records.
4. **System Architecture Diagram:** – Shows how the React frontend, Node.js/Express.js backend, PostgreSQL database, and recommendation service interact.

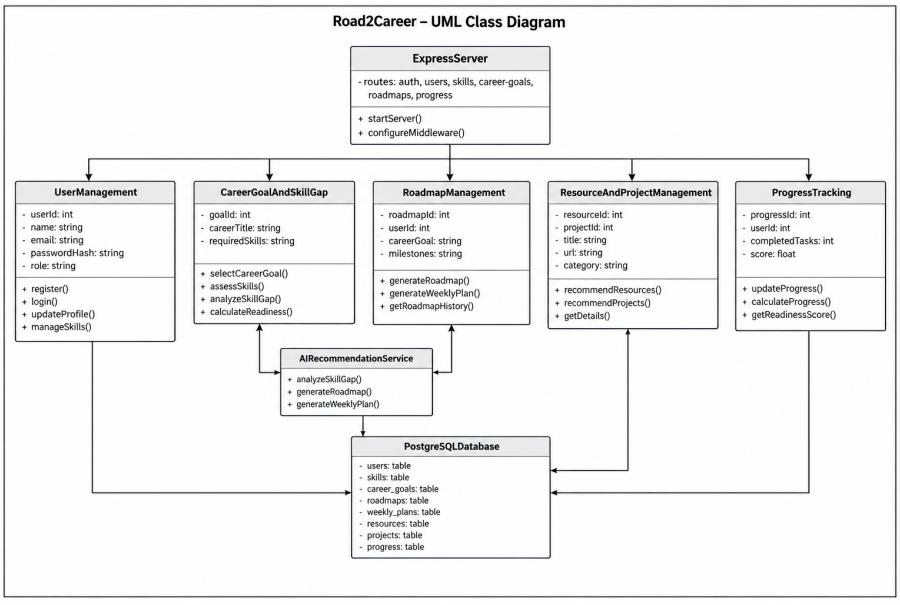
5. Advantages of the Design Methodology:

- Modular Design: Each module can be developed and tested independently.
- Scalability: New career paths, skills, learning resources, and features can be added in the future.
- Personalization: Career recommendations and learning roadmaps are based on student skills and selected career goals.
- Efficiency: Centralized data management and API communication support efficient system operation.
- Security: JWT authentication and password hashing help protect user accounts.
- Maintainability: The layered architecture simplifies debugging, testing, and future modifications.
- Progress Monitoring: Students can monitor learning progress and career readiness through progress-tracking features.

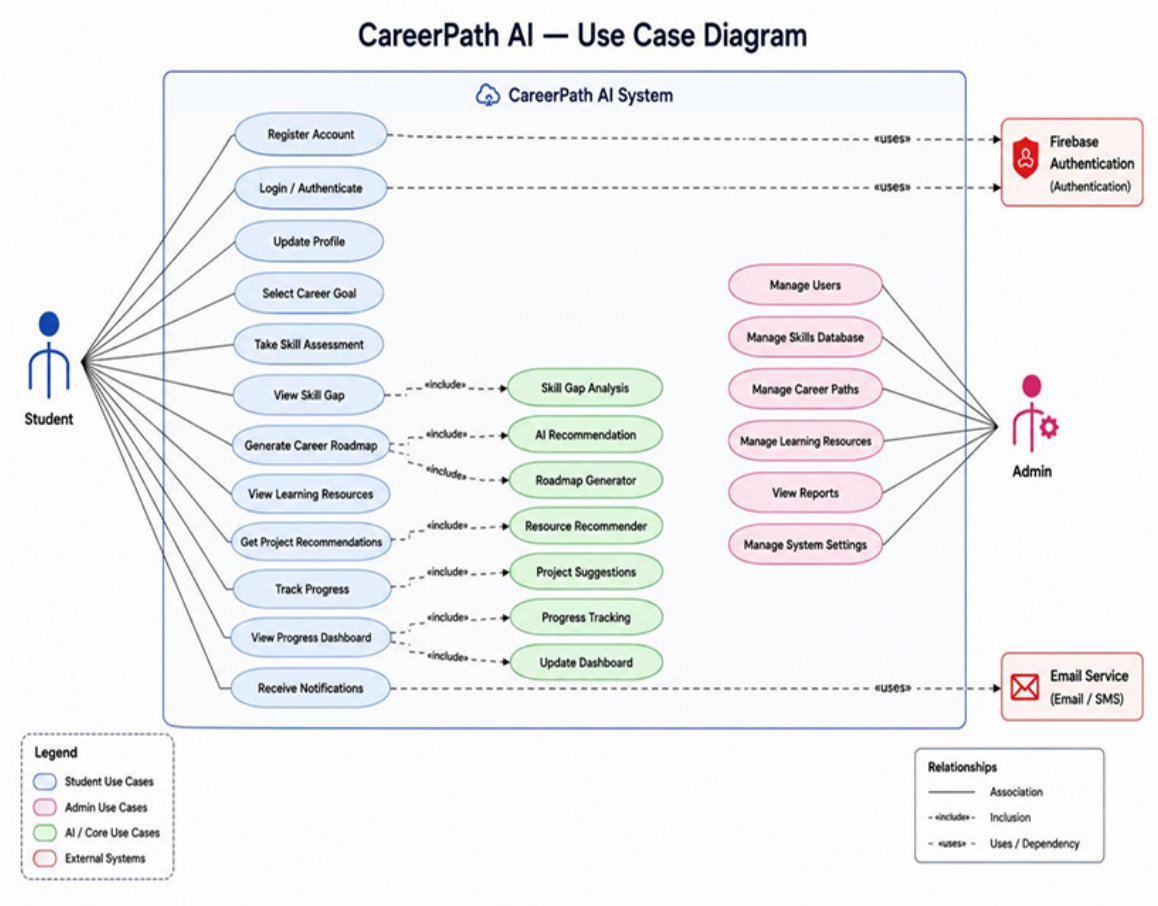
6. Outcome:

The result of this design methodology is a robust, interactive, and scalable Skill Barter Platform that effectively connects learners and teachers, promotes community-based education, and supports continuous learning through gamification and advanced assessments.

3.2 UML diagram



3.3 USE-CASE Diagram



4. AGILE TECHNOLOGY OVERVIEW

Agile is a software development approach that focuses on flexibility, collaboration, and continuous improvement. Unlike the traditional Waterfall model, Agile divides software development into smaller cycles called sprints. Each sprint focuses on developing and testing specific features of the application. The Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator follows an Agile-based development approach to build its modules incrementally. This approach helps organize project tasks, identify issues early, and make improvements throughout development.

Key Principles of Agile:

- Individuals and Interactions: Encourages communication and coordination among team members.
- Working Software: Focuses on developing functional modules of the Road2Career application.
- Customer Collaboration: Considers user requirements and feedback during development.
- Responding to Change: Allows modifications to features and requirements when necessary.

These principles ensure that development remains user-centered and goal-driven, adapting to feedback and evolving requirements. Agile is not a single method but rather a mindset that can be applied in various ways, one of the most popular being Scrum.

4.1 Introduction to Scrum

Scrum is an Agile framework used to organize and manage software development through short, structured iterations called sprints. Each sprint focuses on completing a set of tasks and producing a functional part of the application.

In Road2Career, Scrum-based planning is organized into six sprints, with a planned total of 48 development hours. Each sprint focuses on a particular module of the system.

Roles in Scrum:

1. **Product Owner:**Defines the project requirements, maintains the product backlog, and prioritizes features according to their importance.
2. **Scrum Master:** Facilitates the Scrum process, monitors progress, and helps resolve obstacles during development.
3. **Development Team:** Develops, integrates, and tests the frontend, backend, database, and other project modules.

Key Artifacts in Scrum:

1. **Product Backlog:**Contains the prioritized features and requirements of Road2Career, such as student management, skill-gap analysis, roadmap generation, and progress tracking.
2. **Sprint Backlog:** Contains the tasks selected for each sprint, along with task IDs, descriptions, related user stories, estimated hours, and status.
3. **Increment:** Represents the functional features developed during a sprint, such as student registration, career goal selection, personalized roadmaps, or progress monitoring.

Scrum Ceremonies:

1. **Sprint Planning:**Defines the tasks and goals to be completed during each sprint.
2. **Daily Scrum (Standup):** Reviews completed work, planned activities, and difficulties encountered during development
3. **Sprint Review:** Demonstrates the completed features and checks whether they satisfy the project requirements.
4. **Sprint Retrospective:** Identifies what went well, the problems faced, and improvements that can be made in future sprints.

4.2 Principles or Methodology Used in Scrum

Scrum follows a set of core principles that align with Agile's values but are more specific to the Scrum framework itself:

1. **Commitment:** The development team works toward completing the tasks selected for each sprint.
2. **Focus:** Each sprint concentrates on a specific set of Road2Career features, avoiding unnecessary work.
3. **Openness:** Progress, challenges, and task status are communicated clearly.
4. **Respect:** Team members value each other's contributions and work collaboratively.
5. **Courage:** Team members identify technical issues and communicate problems early.
6. **Continuous Improvement:** Feedback from testing and sprint reviews is used to improve the application and development process.

Scrum also emphasizes empirical process control, which means that decisions are made based on observations of the work being done, allowing for greater flexibility and the ability to adapt to changing conditions.

Methodology:

- **Iterative and Incremental:** Road2Career is divided into six sprints, each focusing on specific modules and features.
- **Self-Organizing Teams:** Development responsibilities and tasks are organized to complete the sprint goals efficiently.
- **Time-Boxed:** Each sprint has a planned duration of 8 development hours, making the total planned development effort 48 hours.
- **Continuous Testing:** Modules are checked during development to identify and correct errors early.
- **Adaptability:** Changes and improvements can be incorporated into later sprints based on testing results and project requirements.

4.3 Product Backlog

ID	User Story / Feature	Priority	Estimation	Status
PB1	Project Setup & Student Module	High	8 pts	Completed
PB2	Career Goal & Skill Gap Analysis	High	8 pts	Completed
PB3	AI Career Roadmap Generation	High	8 pts	Completed
PB4	Project & Learning Resource Module	Medium	8 pts	Completed
PB5	Progress Tracking Module	High	8 pts	Completed
PB6	Admin Module & Final Integration	Medium	8 pts	Completed

4.4 Sprint Planner

Sprint Backlog 1

Product Backlog Item	Sprint ID	Sprint Task	Sprint Goal	Estimated Hours	Acceptance Criteria
As a developer, As a student, I want to register, log in, and manage my profile and skills so that I can securely access the Road2Career platform and receive personalized career recommendations.	1.1	Create React frontend project	Establish the frontend structure.	1	React application starts successfully.
	1.2	Set up Node.js, Express.js, and PostgreSQL	Establish backend and database connectivity.	1	Backend runs and connects to PostgreSQL.

	1.3	Design database schema for users and profiles	Store student information correctly.	1	Required tables and relationships are created.
	1.4	Implement student registration	Allow students to create accounts.	1	Valid student details are saved successfully.
	1.5	Implement student login with JWT authentication	Provide secure login access.	1	Valid login generates a JWT; invalid login is rejected.
	1.6	Develop profile management module	Allow students to update their profiles.	1	Profile changes are saved and displayed.
	1.7	Develop skill management (Add/Edit/Delete Skills)	Allow students to maintain their skills.	2	Students can add, edit, and delete skills.

Sprint Backlog 2

Product Backlog Item	**Sprint ID**	**Sprint Task**	**Sprint Goal**	**Estimated Hours**	**Acceptance Criteria**
As a student, I want to select my career goal so that the system can analyze my skills and identify the missing skills required for my chosen career.	2.1	Create Career Goal Selection page	Allow students to select a career goal.	1	Students can select and save a career goal.
	2.3	Implement Skill Assessment module	Assess students' existing skills.	2	Students can enter or assess their skill levels.

	2.4	Develop Skill Gap Analysis	Identify skills missing for the selected career	2	Missing skills and improvement suggestions are displayed.
	2.5	Calculate Career Readiness Score	Estimate students' readiness for their chosen career.	2	The readiness score is calculated and displayed.

Sprint Backlog 3

Product Backlog Item	**Sprint ID**	**Sprint Task**	**Sprint Goal**	**Estimated Hours**	**Acceptance Criteria**
As a system, I want to generate an AI-powered personalized career roadmap and weekly learning planner based on the student's career goal and skill analysis so that the student has a structured learning path.	3.1	Integrate AI Roadmap Generator	Connect the recommendation service to roadmap generation.	2	The service processes career goals and skill information.
	3.2	Generate Personalized Career Roadmap	Create a learning path based on the student's career goal and skills.	2	A personalized roadmap is generated and displayed.
	3.3	Develop Weekly Learning Planner	Organize learning activities into weekly tasks.	1	Weekly learning tasks are generated and displayed

	3.4	Save Roadmap History	Store generated roadmaps for future access.	1	Students can access previously saved roadmaps.
	3.5	Display Career Timeline	Present roadmap milestones in sequence.	2	Career milestones are displayed in an understandable timeline

Sprint Backlog 4

Product Backlog Item	**Sprint ID**	**Sprint Task**	**Sprint Goal**	**Estimated Hours**	**Acceptance Criteria**
As a system, I want to recommend projects and learning resources based on the student's career goal, skill level, and roadmap so that the student can improve both practical and theoretical knowledge	4.1	Develop Learning Resource Module	Provide learning materials to students.	2	Learning resources are displayed with relevant details.
	4.2	Implement Resource Recommendation System	Recommend resources related to career paths.	2	Relevant resources are categorized and accessible.
	4.3	Develop Project Recommendation Module	Suggest projects for practical learning.	2	Relevant project recommendations are displayed.

	4.4	Create Project Details Page	Display project information.	1	Students can view project descriptions and details.
4.5	Connect Recommendations with Career Roadmap	Link projects and resources to the learning path.	1	Related recommendations are accessible from the roadmap.	

Sprint Backlog 5

Product Backlog Item	Sprint ID	Sprint Task	Sprint Goal	Estimated Hours	Acceptance Criteria
As a student, I want to track my learning progress and career readiness so that I can monitor completed tasks, projects, and overall improvement	5.1	Develop Progress Dashboard	Display student learning progress.	2	Dashboard displays current progress information.
	5.2	Implement Task Completion Tracking	Record completed learning tasks.	1	Completed tasks are recorded correctly..
	5.3	Calculate Progress Percentage	Calculate overall learning progress.	1	Progress percentage updates based on completed tasks.

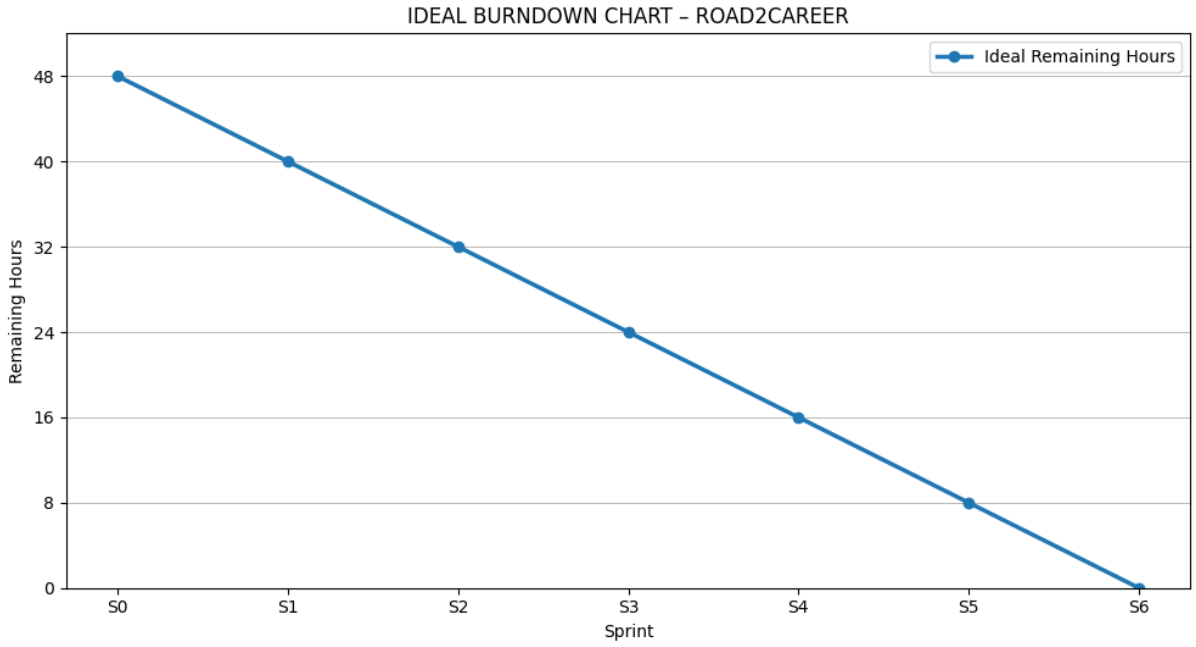
	5.4	Display Learning Statistics	Present learning performance information.	2	Learning statistics are displayed correctly.
	5.5	Implement Achievement Badges	Recognize student learning achievements.	2	Badges are awarded according to defined achievement criteria.

Sprint Backlog 6

Product Backlog Item	**Sprint ID**	**Sprint Task**	**Sprint Goal**	**Estimated Hours**	**Acceptance Criteria**
As an admin, I want to manage students, career paths, skills, projects, learning resources, and view reports so that the platform remains accurate and up to date.	6.1	Develop Admin Dashboard	Provide an interface for administrative activities.	1	Admin dashboard displays the available management options.
	6.2	Implement Student Management	Allow administrators to manage student records.	1	Admin can view and manage student records
	6.3	Implement Career Path and Skill Management	Maintain career paths and skill information.	1	Admin can add, edit, and delete career paths and skills..
	6.4	Implement Project and Resource Management	Maintain project and learning resource records.	1	Admin can manage project and resource information.

	6.5	Generate Reports and Analytics	Provide an overview of system information.	1	Reports and analytics display the available data correctly.
	6.6	System Testing and Bug Fixing	Verify the integrated application and fix identified issues.	2	Major identified errors are corrected and core features are tested.
	6.7	Documentation and Final Deployment	Prepare project documentation and deployment.	1	Documentation is prepared and deployment steps are completed.

4.5 Ideal Burn down chart



5. SYSTEM DESIGN

5.1 Technology / Framework [MVC]

Category	Technology	Version	Description / Purpose
Front-End	JavaScript, HTML, CSS	Latest Stable	Used for structuring, styling, and adding interactivity to the web pages. Forms the base layer of the user interface.
	React.js	18.x (Latest Stable)	Frontend JavaScript library used for creating responsive and dynamic user interfaces following the component-based design.
	Tailwind CSS	5.x / 3.x	UUsed to create responsive layouts and consistent interface styling.
	Vite	Compatible Version	Provides the frontend development server and build tools for the React application.
	Axios	Latest Stable	Handles communication between the frontend and backend through RESTful APIs.
Back-End	Node.js	Installed Version	JavaScript runtime used to execute server-side logic and backend operations.
	Express.js	Installed Version	Manages server-side logic, routing, and API handling as part of the MVC Controller layer..
	PostgreSQL	18	Relational database used to store student data, skills, career goals, roadmaps, and progress records. Represents the data storage layer
	pg Library	Installed Version	Connects the Node.js backend with PostgreSQL for database operations.

Category	Technology	Version	Description / Purpose
	JWT (JSON Web Token)	Latest Stable	Provides secure authentication and user session management.
	bcryptjs	Latest Stable	Hashes passwords securely before storing them in the database.
	Rule-Based Recommendation Service	Project Implementation	Analyzes skill gaps and generates personalized career roadmaps and weekly learning plans.
	Helmet and CORS	Installed Versions	Improves backend security and manages cross-origin requests.
Hosting / Deployment	Node.js-Compatible Web Hosting	As Required	Used to host and deploy the frontend and backend with database connectivity.
Version Control	Git & GitHub	Latest Stable	Manages source code versioning, collaboration, and repository maintenance.

2. Versioning and Software Tools:

Purpose	Tool	Version
Version Control	GitHub	Latest Stable
Code Editor	Visual Studio Code	Latest Stable
Implementation	React.js (Frontend), Node.js and Express.js (Backend)	Latest Versions
Database Management	PostgreSQL	Version 18
API Testing	Postman	Latest Stable
Database Connectivity	pg Node.js Library	Installed Project Version

Purpose	**Tool**	**Version**
Frontend Development	Vite	Installed Project Version
Browser Testing	Google Chrome, Microsoft Edge	Latest Stables

5.2 Module Description

Module Name	**Description**
Student Management Module	Handles student registration, login, JWT authentication, profile management, and skill management. Allows students to securely access the system and maintain their personal and skill details.
Career Goal and Skill Gap Analysis Module	Allows students to select their career goals and assess their existing skills. Compares current skills with the skills required for the selected career, identifies missing skills, and calculates the career readiness score..
Personalized Career Roadmap Module	Generates personalized career roadmaps based on the selected career goal, existing skills, and identified skill gaps. Provides a structured learning path, weekly learning plans, career timelines, and roadmap history.
Project and Learning Resource Module	Recommends relevant learning resources and practical projects based on the student's career goals and skill requirements. Helps students improve their knowledge and gain practical experience.
Progress Tracking Module	Tracks completed learning tasks and monitors student progress. Displays progress percentages, learning statistics, and career readiness information to help students understand their development.
Admin Management Module	Provides administrative functions for managing student records, career paths, skills, projects, and learning resources. Supports system monitoring and report generation where implemented.

6. CODING

6.1 Coding Standards Used in the Project

To ensure readability, consistency, and maintainability, the development of Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator follows standard coding practices. These practices help reduce errors, improve code organization, and simplify future modifications.

1. General Coding Practices

- Code is organized into modular and reusable components.
- Meaningful and descriptive names are used for variables, functions, and files
- Consistent indentation, spacing, and formatting are maintained throughout the project.
- Comments are used where necessary to explain complex logic and important functions.
- Git and GitHub are used to maintain source code history and track changes. Frontend, backend, and database-related code are organized separately

2. Front-End (React.js, HTML, CSS, JavaScript) Standards

- HTML5 standards are followed for structuring web pages.
- React components use PascalCase naming conventions (e.g., StudentProfile.jsx).
- JavaScript variables and functions use camelCase naming conventions (e.g., careerGoal, generateRoadmap()).
- React components are designed to be reusable and easy to maintain.
- Tailwind CSS classes are used to create responsive layouts and consistent styling.
- React Router DOM is used to manage navigation between pages.
- Axios is used to communicate with backend APIs.
- Form inputs and user interactions are designed to provide clear feedback and validation.

3. Back-End (Node.js and Express.js) Standards

- JavaScript naming conventions are followed, using camelCase for variables and functions.
- Express.js routes are organized according to their respective modules.
- Controllers handle request processing and business logic.
- Middleware is used for authentication, validation, and error handling where implemented.
- RESTful API principles are followed for communication between the frontend and back-end.
- JWT is used to authenticate users and protect authorized routes.
- The rule-based recommendation service is organized into functions for skill-gap analysis, roadmap generation, and weekly learning plan generation.
- Appropriate error handling is used to manage invalid requests and server-side failures.

4. Database Standards (PostgreSQL)

- Database tables and columns use consistent naming conventions.
- Primary keys are used to uniquely identify database records.
- Foreign keys are used to establish relationships between related tables where required.
- Data types are selected according to the information being stored.
- SQL queries are used to insert, retrieve, update, and delete records.
- Input validation and database constraints help maintain data integrity.
- Database connection settings are maintained separately from application logic.

5. Security and Quality Standards

- Passwords are hashed using bcryptjs before storage.
- JWT authentication is used to manage access to protected application features.
- Input validation is applied to reduce invalid or unexpected data.
- Helmet is used to improve HTTP security headers.
- CORS settings control cross-origin requests.
- Sensitive configuration values are stored in environment variables rather than hard-coded into source files.
- API endpoints are tested to verify correct responses and error handling.

6. Code Review and Testing

- Git commits are used to track code modifications.
- Frontend components are checked for correct rendering and navigation.
- Backend APIs are tested for correct request and response handling.
- Database operations are checked to ensure data is stored and retrieved correctly.
- Authentication and authorization are tested for protected routes.
- Integration testing verifies communication between the frontend, backend, and PostgreSQL database.
- Identified errors are corrected to improve system reliability.

2. HTML Frontend

Structure and Semantics
a. HTML5 Doctype:
Starts with a proper `<!DOCTYPE html>`, ensuring the browser renders the page in standards mode.
b. Semantic Tagging:
Uses appropriate HTML tags such as `<head>`, `<body>`, `<table>`, `<tr>`, `<td>`, and `<button>` for logical content grouping.
c. Descriptive UI Elements:
Buttons like Start and Stop are labeled clearly, contributing to user-friendly interaction.

Styling and Layout
a. Basic Layout Consistency:
The page is structured for straightforward navigation, with content grouped into appropriate sections (e.g., table layout for displaying network traffic).
b. Expandable Design:
Layout allows easy integration of additional features such as popups, color-coded threats, and notification banners.

Accessibility and Maintainability
a. JavaScript Integration:
JavaScript (inline or external) is used to dynamically populate table data and handle button interactions. The project benefits from this by offering real-time monitoring experiences.
b. Separation of Concerns (recommended improvement):
Inline JavaScript can be moved to a separate .js file for better modularity and reusability.
c. Accessibility Considerations (recommended improvement):
Add role, aria-label, and tabindex attributes for screen reader compatibility.

3. General Coding Standards and Best Practices

Code Readability

- Clean, concise logic that's easy to understand even for developers who are new to the project.
- Proper spacing and indentation throughout both files.

Reusability and Extensibility
Design permits the addition of more modules like threat detection, logging, or advanced UI components without disrupting the existing flow.

Maintainability

- Follows a clean separation between server-side (Flask) and client-side (HTML/JS), which eases debugging and future updates.

Error Handling and Validation (recommended for future)

- Implement try-except blocks in Python to handle API failures gracefully.
- Form validations and user input sanitation can be added if interactive features are planned.

Documentation and Comments (to be improved)

- In-code comments, especially above function definitions, would greatly enhance clarity.
- A README.md or inline docstrings should be included to help onboard new developers or explain the purpose of the project modules.

Summary

Area	**Followed standard**	**Recommended improvements**
JavaScript Naming	camelCase	Maintains consistent naming for variables and functions.
React Components	PascalCase	Improves component identification and readability.
Frontend Design	React.js and Tailwind CSS	Supports reusable and responsive interfaces.
Backend Design	Node.js and Express.js	Organizes APIs, controllers, and business logic.
Database	PostgreSQL	Maintains structured data and relationships.
Authentication	JWT and bcryptjs	Supports authentication and password security.
API Design	RESTful principles	Provides structured frontend-backend communication.
Error Handling	Validation and error middleware	Helps manage invalid requests and failures.
Version Control	Git and GitHub	Tracks changes and supports code maintenance.
Documentation	README and code comments	Helps developers understand and maintain the system.

6. TESTING

6.1 Unit Testing – Test Cases

Test Case ID	Module	Test Scenario	Input	Expected Output	Actual Output	Status
TC001	Student Management	Student Registration	Valid student details submitted through registration form	Student details saved in PostgreSQL and success message displayed	As expected	Pass
TC002	Authentication	Student Login	Valid email and password	JWT token returned and user redirected to dashboard	As expected	Pass
TC003	Authentication	Invalid Login	Incorrect email or password	Error message displayed and access denied	As expected	Pass
TC004	Profile Management	View Student Profile	Authenticated request to profile API	Student profile details displayed correctly	As expected	Pass
TC005	Skill Management	Add Student Skill	Skill name and proficiency level	Skill saved and displayed in the student's profile	As expected	Pass
TC006	Skill Management	Update or Delete Skill	Existing skill details	Skill updated or removed successfully	As expected	Pass
TC007	Career Goal	Select Career Goal	Valid career goal selected by student	Career goal saved successfully	As expected	Pass
TC008	Skill Gap Analysis	Analyze Missing Skills	Selected career goal and existing student skills	Missing skills identified based on career requirements	As expected	Pass
TC009	Career Readiness	Calculate Readiness Score	Student skills, proficiency, projects, and consistency data as applicable	Readiness score calculated and displayedn	As expected	Pass

Test Case ID	Module	Test Scenario	Input	Expected Output	Actual Output	Status
TC010	Career Roadmap	Generate Personalized Roadmap	Selected career goal and skill gap information	Career roadmap generated according to the implemented recommendation logic	As expected	Pass
TC011	Weekly Learning Planner	Generate Weekly Plan	Career roadmap and learning requirements	Weekly learning activities displayed	As expected	Pass
TC012	Roadmap History	Save and Retrieve Roadmap	Generated roadmap and authenticated student account	Roadmap history saved and retrieved correctly	As expected	Pass
TC013	Learning Resources	View Recommended Resources	Selected career goal or required skill	Relevant learning resources displayed	As expected	Pass
TC014	Project Recommendation	View Recommended Projects	Career goal and skill requirements	Relevant practical projects displayed	As expected	Pass
TC015	Progress Tracking	Mark Learning Task Complete	Student selects a task as completed	Completion status and progress updated correctly	As expected	Pass
TC016	Progress Tracking	View Progress Dashboard	Student progress records	Progress percentage and learning statistics displayed correctly	As expected	Pass
TC017	Access Control	Access Protected API Without Token	API request without a valid JWT	Request rejected with an unauthorized response	As expected	Pass
TC018	Admin Management	Manage Career Paths and Skills	Authorized admin request	Career paths and skill records managed according to permissions	As expected	Pass

7. IMPLEMENTATION

The Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator project was developed using an incremental development approach based on Agile principles. The system was divided into six sprints, with each sprint focusing on specific modules and functionalities. This approach helped organize development, integrate the frontend and backend, and identify errors during implementation.

Requirement Analysis and Planning

Requirement analysis was carried out to identify the functional and non-functional requirements of the Road2Career system.

- Functional requirements included student registration and login, profile and skill management, career goal selection, skill gap analysis, personalized career roadmap generation, weekly learning planning, project and learning resource recommendations, progress tracking, and admin management.
- Non-functional requirements included security, usability, maintainability, reliability, and responsive user interface design.
- A product backlog was prepared to organize the project requirements into user stories and development tasks.
- The project was divided into six sprints, with a planned total of 48 development hours and 8 hours allocated to each sprint.

Time estimates and sprint goals were aligned with stakeholder expectations and available development effort.

Modular Design and Development

The system was developed using React.js for the frontend, Node.js and Express.js for the backend, and PostgreSQL for database management. The application follows a modular structure in which each module handles specific functions.

The primary modules developed include:

Core Modules Developed:

- **Student Management Module**
  - Provides student registration and login.
  - Supports profile management and skill management.
  - Uses JWT-based authentication for protected operations.
- **Career Goal and Skill Gap Analysis Modulet**
  - Allows students to select their desired career path.
  - Compares existing skills with the skills required for the selected career.
  - Identifies missing skills and supports career readiness assessment using the implemented scoring logic.
- **Personalized Career Roadmap Module**
  - Generates a career roadmap based on the selected career goal and identified skill gaps.

  - Organizes learning activities into a structured sequence.
  - Supports weekly learning plans, career timelines, and roadmap history where implemented.

- **Project and Learning Resource Module**
  - Provides relevant projects and learning resources to support skill development.
  - Helps students gain practical knowledge related to their selected career.
  - Connects recommended learning activities with career development goals.

- **Progress Tracking Module**
  - Tracks completed learning tasks.
  - Displays progress percentages and learning statistics.
  - Helps students monitor their progress toward career goals.

- **Admin Management Module**
  - Supports administrative management of students, career paths, skills, projects, and learning resources.
  - Provides administrative views and reporting functions where implemented.t.

Testing and Debugging

Unit Testing

- Individual functions and modules were checked to verify their expected behavior.
- Registration and login operations were checked with valid and invalid inputs.
- Skill gap analysis and roadmap generation logic were checked using different career goals and skill combinations.
- Progress calculation and task completion behavior were checked against the expected results..

Integration Testing

- Frontend and backend communication was checked through API requests.
- Database operations were checked to verify that student information and related records could be stored and retrieved.
- Authentication was checked when accessing protected API endpoints.
- Integration between career goal selection, skill gap analysis, roadmap generation, and progress tracking was checked.

Debugging

- Browser developer tools were used to identify frontend errors and interface problems.
- Server logs and terminal output were used to investigate backend errors.
- API requests and responses were checked to identify communication problems.
- PostgreSQL connection settings and database queries were checked when database errors occurred.

- Authentication tokens, input validation, and error-handling logic were reviewed to resolve related issues.

Deployment

- The React.js frontend was configured to communicate with the Node.js and Express.js backend.
- PostgreSQL was used to store application data.
- The backend API endpoints were checked before integrating them with the frontend.
- Environment configuration was used to manage database connections and other application settings.
- Git and GitHub were used for source code version control and repository management.
- The application can be deployed using hosting services compatible with React, Node.js, and PostgreSQL

Maintenance and Continuous Improvement

- - Errors identified during testing can be corrected through updates to the relevant modules.
- Database queries and API responses can be optimized to improve performance.
- The recommendation logic can be refined to improve the relevance of career roadmaps and learning plans.
- Additional career paths, skills, projects, and learning resources can be added as needed.
- Future enhancements may include advanced career recommendations, improved analytics, resume assessment, notifications, and more detailed learning progress reports.

8. BUSINESS PERSPECTIVE

The implementation of Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator provides strategic, educational, and economic value by helping students make informed career decisions and develop the skills required for their chosen career paths. In today's competitive job market, students often face difficulties identifying suitable careers, understanding skill requirements, and planning their learning activities. Road2Career addresses these challenges by providing a centralized platform for career goal selection, skill gap analysis, personalized career roadmaps, learning resource recommendations, and progress tracking. By using modern web technologies and a rule-based recommendation approach, the system offers a structured and user-friendly solution for career development.

Strategic Alignment and Market Relevance

- The platform aligns with the growing demand for digital career guidance, personalized learning, and skill-based development.
- Road2Career supports students in identifying suitable career paths and understanding the skills required to achieve their professional goals.
- It addresses the need for structured learning plans by combining career guidance, skill gap analysis, and progress monitoring in one platform.
- The system can be extended to support different academic backgrounds, career fields, and changing industry skill requirements.
- Its modular architecture allows future integration of advanced AI technologies and additional career development features.

Value Proposition

Real-Time Visibility: Enables businesses to view, analyze, and act upon live network activity, reducing incident response time.

- Personalized Career Guidance: Helps students explore career paths based on their selected goals, existing skills, and learning needs.
- Skill Gap Analysis: Identifies missing skills by comparing a student's current skills with the skills required for a selected career.
- Personalized Career Roadmap: Provides a structured learning path that helps students understand the steps required to prepare for their target career.
- Weekly Learning Planner: Organizes learning activities into manageable tasks to encourage consistent skill development.
- Project and Resource Recommendations: Helps students discover relevant learning resources and practical projects to improve their knowledge.
- Progress Tracking: Enables students to monitor task completion, learning progress, and career readiness.
- Cost-Effective Solution: Uses open-source technologies, reducing software licensing costs and making the platform suitable for academic and small-scale deployment.

Operational Impact

- Empowering Students: Helps students make informed career decisions and follow a clear learning path.
- Improved Learning Efficiency: Organizes career goals, skill requirements, learning resources, and tasks in a centralized platform.
- Better Resource Utilization: Reduces the need to switch between multiple platforms to manage career preparation activities.
- Progress Monitoring: Allows students to review completed tasks and identify areas that require further improvement.
- Administrative Efficiency: Provides an administrative module for managing student information and relevant career-related data, according to the implemented features.
- Data-Informed Improvement: Progress and skill information can help guide future improvements to recommendations and learning plans..

Competitive Advantage

- Integrated Career Development: Combines career goal selection, skill gap analysis, roadmap generation, and progress tracking in one system.
- Personalized Learning Approach: Focuses on individual career goals and existing skills rather than providing only general career information.
- Structured Roadmaps: Helps students break long-term career objectives into smaller, manageable learning activities.
- Practical Skill Development: Connects career preparation with learning resources and project recommendations.
- Affordable Development: Uses technologies such as React.js, Node.js, Express.js, and PostgreSQL, which are widely available and supported by open-source communities.
- Extensibility: The modular design supports future improvements without requiring the entire system to be redesigned.

Future Scalability and Commercial Potential

- Expansion to Mobile Platforms: The system can be extended into Android and iOS applications to improve accessibility.
- Advanced AI Recommendations: Machine learning techniques can be integrated in future versions to provide more sophisticated career and learning recommendations.
- Institutional Adoption: Colleges and training institutions could use the platform to support student career planning and skill development.
- Subscription-Based Model: A future commercial version could offer premium career assessments, advanced roadmaps, and detailed progress reports.
- Industry Collaboration: Partnerships with companies and training providers could help keep career requirements, learning resources, and project suggestions relevant.
- Resume and Interview Preparation: Additional modules could provide resume guidance, interview practice, and job-readiness assessments.

Risk Mitigation

- Data Security: JWT-based authentication, password hashing with bcryptjs, and appropriate backend security controls help protect user accounts.
- Data Integrity: PostgreSQL provides structured data storage and supports relationships between users, skills, career goals, and learning progress.
- System Reliability: Input validation, error handling, logging, and appropriate API testing can help identify and manage system failures.
- Recommendation Accuracy: The rule-based recommendation service can be refined and evaluated against different career goals and skill profiles.
- Project Management: Sprint-based development helps organize implementation into manageable tasks and makes it easier to identify delays.
- Continuous Improvement: User feedback, testing, and updates can help improve usability, functionality, and recommendation quality over time.

9. FUTURE SCOPE OR ENHANCEMENTS

The current version of Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator provides a foundation for helping students choose suitable career paths, identify skill gaps, generate personalized learning roadmaps, and track their progress. However, several opportunities exist to improve the system's intelligence, accessibility, accuracy, and functionality. The following enhancements can make Road2Career a more advanced and comprehensive career development platform.

Advanced AI-Based Career Recommendation System

- Integrate machine learning algorithms to recommend suitable career paths based on students' skills, interests, academic backgrounds, and career goals.
- Improve the recommendation system using student feedback, learning history, and career preferences.
- Develop an intelligent career prediction feature to suggest suitable job roles based on individual skill profiles.
- Generate more adaptive learning roadmaps based on students' progress, proficiency levels, and changing career requirements

Advanced Skill Assessment and Gap Analysis

- Introduce adaptive assessments that adjust question difficulty according to student performance.
- Provide coding tests and practical assignments to evaluate programming and technical skills.
- Generate detailed skill assessment reports highlighting strengths, weaknesses, and areas for improvement.
- Update skill gap analysis dynamically as students acquire new skills and complete learning activities.

Intelligent Learning and Roadmap Management

- Introduce dynamic roadmap updates based on completed tasks, assessment results, and learning progress.
- Generate personalized daily and weekly study schedules according to students' available learning time.
- Provide estimated completion dates for career roadmaps based on the student's learning pace.
- Add reminders and notifications for upcoming tasks, incomplete activities, and learning milestones.
- Allow students to compare previous and current roadmaps to understand their development over time.

GitHub-Based Project Evaluation

- Integrate GitHub API services to validate submitted repository URLs and check repository availability.
- Analyze repository files and metadata to identify relevant programming, machine learning, and data science projects.
- Verify whether a submitted project contains relevant source code and supporting files before marking a learning task as complete.
- Provide feedback on project submissions and suggest improvements based on predefined evaluation criteria.
- Award points and update progress after successful project validation.

Advanced Progress Tracking and Gamification

- Introduce achievement badges, learning points, and milestones to motivate students.
- Develop leaderboards to encourage healthy competition among learners.
- Provide detailed dashboards displaying skill development, completed tasks, learning consistency, and career readiness.
- Generate weekly and monthly progress reports to help students evaluate their performance.
- Introduce personalized recommendations for improving progress when students fall behind their learning schedules.

Resume Building and Interview Preparation

- Develop an automated resume builder based on students' education, skills, projects, and achievements.
- Provide resume suggestions based on selected job roles and required skills.
- Introduce mock interview questions for technical and HR interview preparation.
- Evaluate practice answers using predefined criteria and provide improvement suggestions.
- Recommend additional skills and projects that can improve students' job readiness.

Mobile Application Integration

- Develop Android and iOS applications using technologies such as React Native or Flutter.
- Provide push notifications for learning reminders, assessment results, and roadmap updates.
- Enable students to access their career roadmaps and progress dashboards from mobile devices.
- Introduce offline access to saved roadmaps and selected learning materials where supported.

Cloud Integration and Scalability

- Deploy the application on cloud platforms such as AWS, Microsoft Azure, or Google Cloud.

- Configure scalable hosting to support an increasing number of students and administrators.
- Implement automated database backups and recovery mechanisms to protect user information.
- Improve API performance and database query efficiency as the volume of users and learning data grows.

Advanced Security and Authentication

- Introduce two-factor authentication (2FA) for additional account security.
- Strengthen role-based access control for students and administrators.
- Implement secure password recovery and email verification features.
- Improve input validation, API protection, and monitoring of suspicious activities.
- Apply appropriate data privacy practices to protect student profiles, assessment results, and learning records..

Integration with External Platforms

- Integrate learning platforms and educational resource providers to offer a wider range of courses and learning materials.
- Connect with job portals and career websites to provide relevant internship and job opportunities.
- Integrate professional platforms such as LinkedIn to support career profile development, where permitted.
- Provide APIs for educational institutions to integrate Road2Career into their existing student support systems.
- Keep career skill requirements and learning resources updated using reliable external data sources

This future roadmap ensures that the Profexia Skill Barter Platform evolves from a functional learning exchange system into a comprehensive, AI-powered, and globally adaptive learning ecosystem — capable of transforming the way people acquire, share, and validate skills in the digital age.

10. APPENDIX

10.1. Git History [Sprint Wise]

Sprint 1: Project Initialization and Backend Setup

- **Commit:** initial-project-setup
  - Initialized the React frontend and Node.js/Express.js backend project structure.
  - Configured the development environment and Git version control.
- **Commit:** postgresql-database-setup
  - Connected the project to MongoDB (Atlas/Local).
  - Created base collections for users, courses, and sessions.
- **Commit:** auth-module-complete
  - Configured the PostgreSQL database connection.
  - Created the initial database schema for student profiles and related information.
- **Commit:** student-authentication-module
  - Implemented student registration and login functionality.
  - Integrated JWT-based authentication and password hashing.

Sprint 2: Career Goal and Skill Gap Analysis

- **Commit:** careergoal-selection
  - Developed the interface for selecting career goals.
  - Connected the frontend with the backend career goal APIs.
- **Commit:** skill-assessment-module
  - Implemented skill management and assessment functionality.
  - Added functionality to compare existing skills with career requirements.
- **Commit:** skill-gap-analysis
  - Developed the rule-based skill gap analysis functionality.
  - Added support for identifying missing skills and assessing career readiness

Sprint 3: Personalized Career Roadmap Generation

- **Commit:** career-roadmap-generation
  - Implemented the career roadmap generation functionality.
  - Used career paths and skill information to prepare structured learning plans.
- **Commit:** weekly-learning-planner
  - Developed the weekly learning planner to organize learning activities.
  - Arranged learning tasks into a structured schedule.
- **Commit:** roadmap-history-and-timeline

  - Added functionality for accessing saved roadmap information and displaying career progress timelines, where implemented.

Sprint 4: Project and Learning Resource Module

- **Commit:** learning-resource-module
  - Developed functionality for displaying relevant learning resources.
  - Organized resources to support students' career goals.
- **Commit:** project-recommendation-feature
  - Implemented project recommendations related to career paths and required skills.
  - Added project information to support practical learning.
- **Commit:** roadmap-resource-integration
  - Connected relevant projects and learning resources with the career preparation process.

Sprint 5: Progress Tracking Module

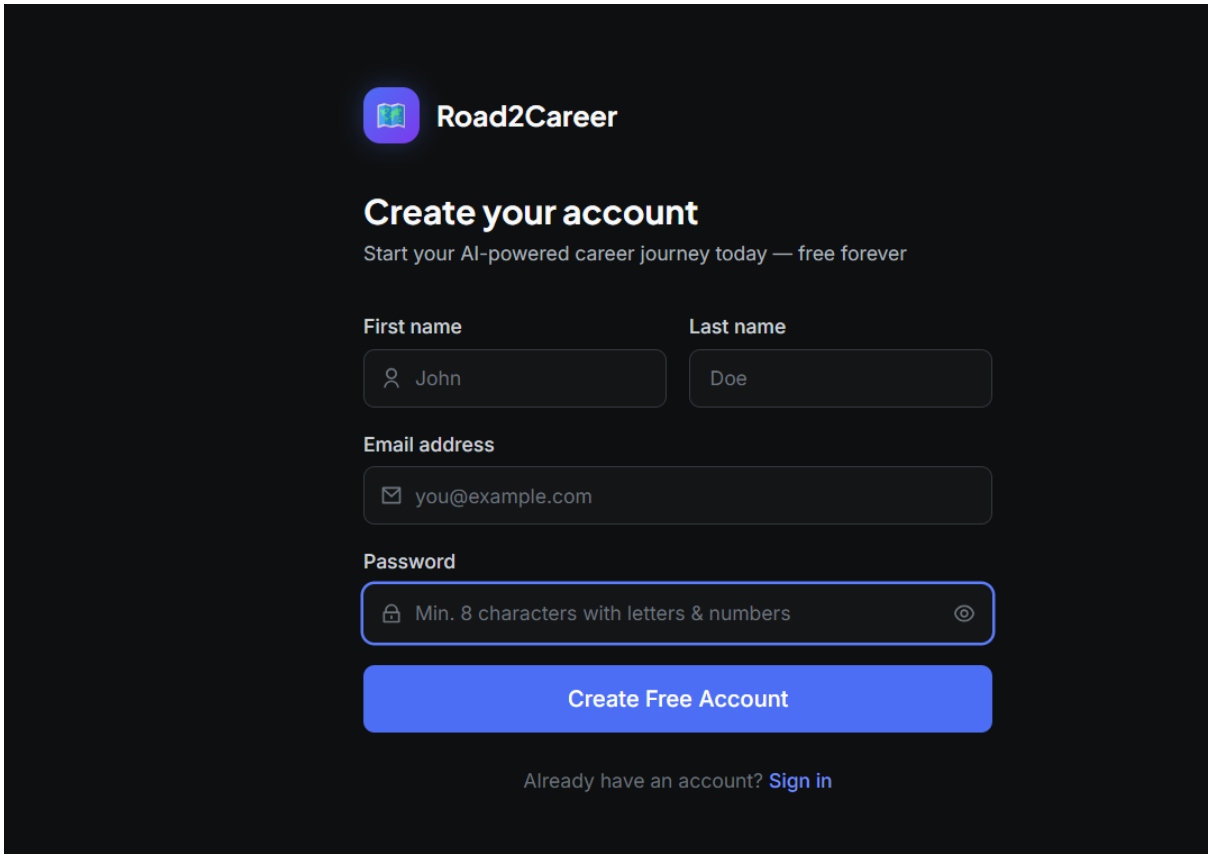
- **Commit:** progress-dashboard-development
  - Developed the progress dashboard to display learning progress.
  - Organized completed and pending learning activities
- **Commit:** task-completion-tracking
  - Implemented functionality for tracking completed learning tasks.
  - Updated progress information based on task completion.
- **Commit:** career-readiness-and-learning-statistics
  - Integrated career readiness and learning statistics features.
  - Improved the visibility of student progress and skill development.

Sprint 6: Admin Module and System Integration

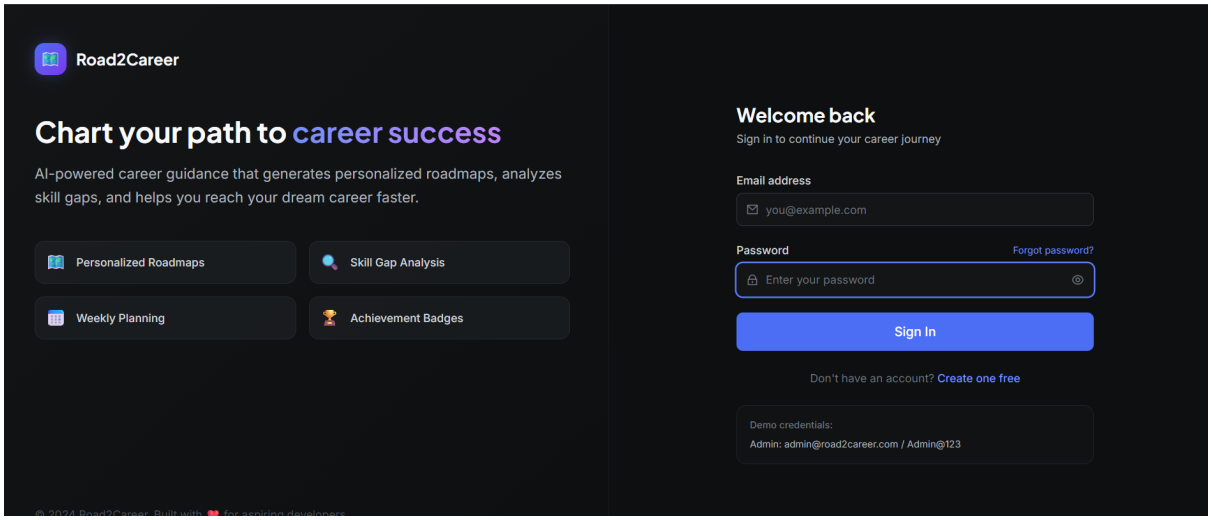
- **Commit:** admin-dashboard-development
  - Developed the administrative interface for managing relevant system information.
  - Added administrative functions for student and career-related data, where implemented.
- **Commit:** api-integration-and-error-handling
  - Integrated frontend components with backend APIs.
  - Improved request validation and error handling.
- **Commit:**system-testing-and-documentation
  - Prepared the application for testing and reviewed the integrated modules.
  - Updated project documentation and supporting materials.

10.2. Screenshots of Forms, Reports, Diagrams (Burn Down Charts)

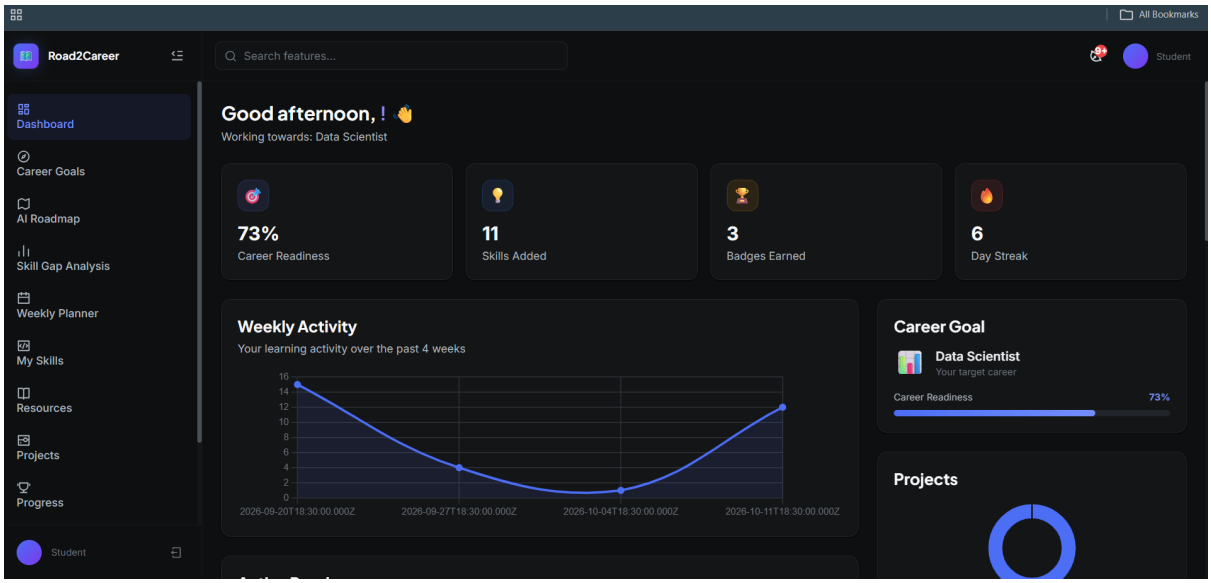
Sign-up form



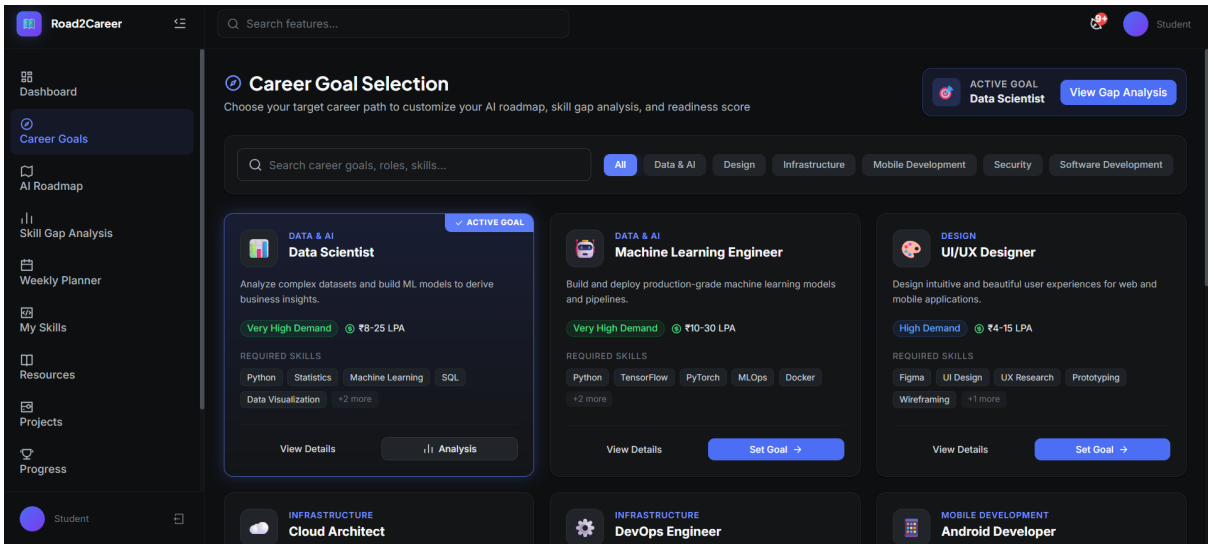
Log-in form



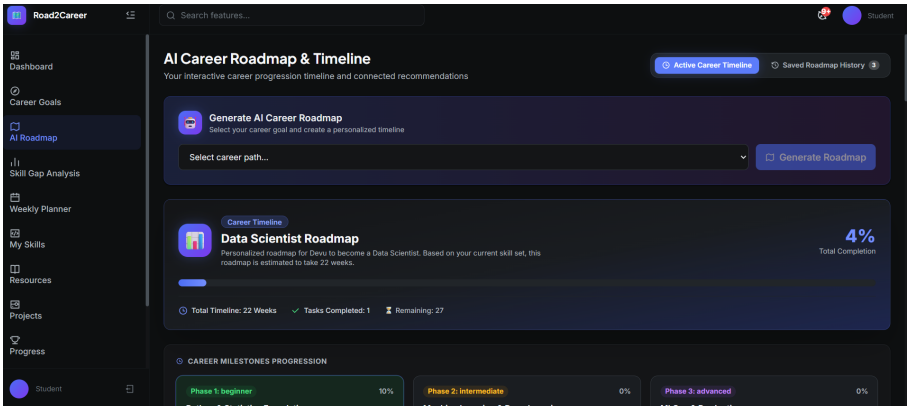
Student Dashboard



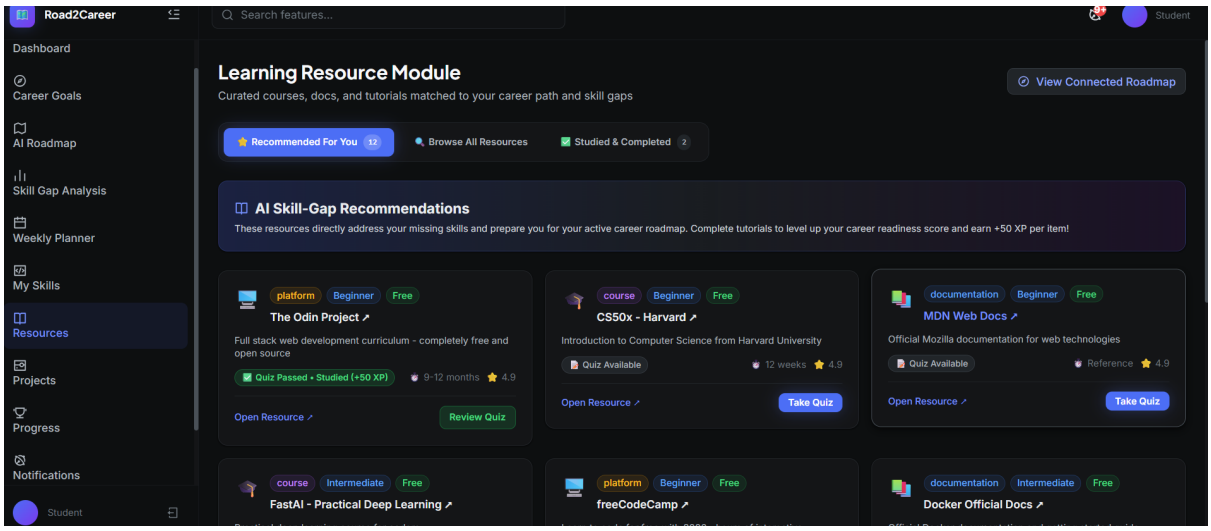
career Dashboard



AI Roadmap



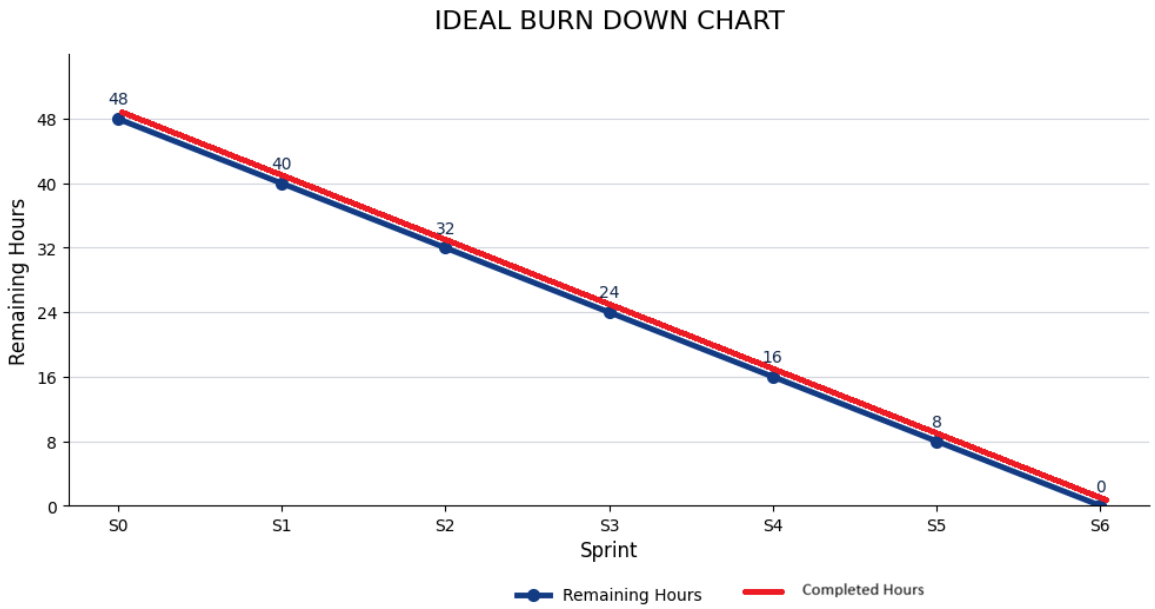
Resourse



Admin page



11.3 Burndown Chart



11. References / Bibliography

Research Papers and Articles

1. 1. Dhanalakshmi, R., Navanidhan, V., Barath, V., Hemanth, V. (2025). AI-Powered Career Navigator: Personalized Roadmap. International Journal of Creative Research Thoughts (IJCRT), Volume 13, Issue 11, ISSN: 2320-2882.
2. 2. Priskilla Angel Rani, J., Selva Lakshmi, D., Sudharshini, S., Ragavi Muthu Thangam, R. (2025). Personalized Career Roadmap Generation. International Journal of Novel Research and Development (IJNRD), Volume 10, Issue 4, ISSN: 2456-4184.
3. 3. ETV Bharat. (2026, July 15). CUSAT Researchers Secure Patent for AI-Based Aptitude Mapping and Career Guidance System. Available at: https://www.etvbharat.com/en/bharat/cusat-researchers-secure-patent-for-ai-based-aptitude-mapping-and-career-guidance-system-enn26071504937

Research Papers

1. Singh, A., & Sharma, M. (2021). Enhancing E-Learning Platforms Using Collaborative Learning Models. International Journal of Computer Applications, 175(12), 23–29.
2. Patel, R., & Deshmukh, P. (2022). A Study on Skill Sharing Platforms Using Web Technologies. Journal of Emerging Technologies and Innovative Research (JETIR), 9(5), 1021–1028.
3. Chen, L., & Zhang, J. (2020). AI-Driven Education Platforms for Personalized Learning. IEEE Transactions on Learning Technologies, 13(4), 743–754.

Official Websites and Framework Documentation

1. React.js Documentation – https://react.dev — Reference for building user interfaces using React.
2. Node.js Documentation – https://nodejs.org/docs/ — Reference for server-side JavaScript development.
3. Express.js Documentation – https://expressjs.com/ — Reference for developing backend APIs.
4. PostgreSQL Documentation – https://www.postgresql.org/docs/ — Reference for relational database design, SQL queries, and database management.
5. Axios Documentation – https://axios-http.com/ — Reference for HTTP requests and frontend-backend communication.
6. JSON Web Token (JWT) – https://jwt.io/ — Reference for token-based authentication.
7. React Router Documentation – https://reactrouter.com/ — Reference for navigation and routing in React applications.

Technical Documentation

1. PostgreSQL Node.js Driver (pg) – https://node-postgres.com/ — Reference for connecting Node.js applications to PostgreSQL.
2. Git Documentation – https://git-scm.com/doc — Reference for source code version control.

3. GitHub Documentation – https://docs.github.com/ — Reference for repository management and collaboration.
4. Winston Documentation – https://github.com/winstonjs/winston — Reference for application logging.

Online Resources

1. Stack Overflow – https://stackoverflow.com/ — Community discussions for troubleshooting programming and integration issues.
2. GitHub – https://github.com/ — Source code hosting and version control resources.
3. GeeksforGeeks – https://www.geeksforgeeks.org/ — Programming, data structures, algorithms, and database learning resources.
4. MDN Web Docs – https://developer.mozilla.org/ — Reference for JavaScript, HTML, CSS, and web development.
5. freeCodeCamp – https://www.freecodecamp.org/ — Tutorials and practical exercises for web development.

Standards and Guidelines

1. OWASP Foundation. OWASP Top 10: Web Application Security Risks. https://owasp.org/www-project-top-ten/
2. World Wide Web Consortium (W3C). Web Standards. https://www.w3.org/standards/
3. Representational State Transfer (REST) API Design Reference. https://restfulapi.net/

12.CONCLUSION

The Road2Career – AI-Powered Personalized Career Mentor and Roadmap Generator is designed to help students make informed career decisions by providing a structured and personalized approach to career planning. The system brings important career development activities together in one platform, including student profile management, career goal selection, skill gap analysis, personalized career roadmap generation, weekly learning plans, project and learning resource recommendations, and progress tracking. The project uses React.js for the frontend, Node.js and Express.js for the backend, and PostgreSQL for database management. A rule-based recommendation service helps analyze students' skills and generate suitable learning roadmaps based on their selected career goals and skill requirements. This approach aims to reduce the confusion students may face when choosing a career and identifying the skills they need to develop. Overall, Road2Career provides a centralized platform that supports students in planning their learning activities, improving their skills, and monitoring their career preparation progress. Future enhancements may include advanced machine learning-based recommendations, automated project evaluation, resume analysis, and interview preparation features. Thus, the project provides a foundation for a more accessible, organized, and personalized career guidance system.